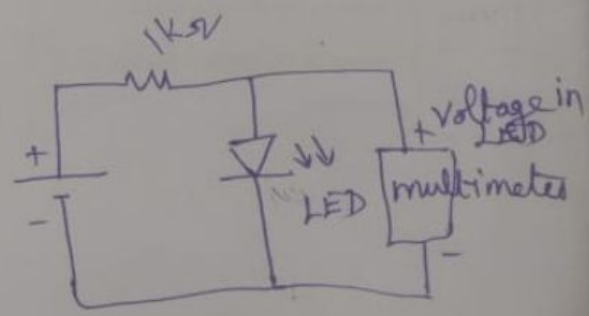
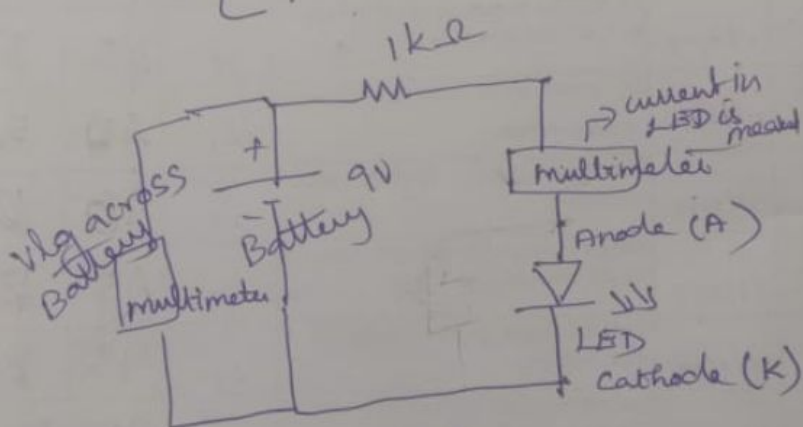


EX.NO:	1
DATE:	10/12/25

To measure the voltage
Components Required:

S. No	Components
01	LED
02	Resistance (10K)
03	Battery (9V)
04	Bread board and connecting wires

Exp: 1
Basic ckt -
Connect like this & measure



try:
When a resistor is connected by restricting the flow of current all change within voltage surge also. So a series resistor

An LED has two terminals the negative is known as the cathode. The two arrows outwards. The longer terminal is the anode and the shorter term is the cathode and they are arranged in any way so

The polarity of batteries positive & negative will be indicated by the '+' sign. The direction of this circuit can be determined by the direction of current flow. The resistor is connected in series

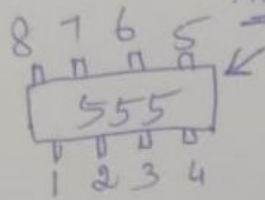
0.01, 0.1

0010 0011

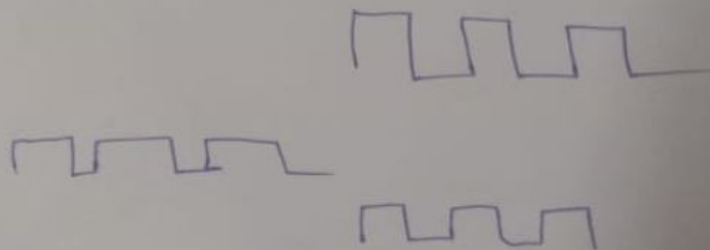
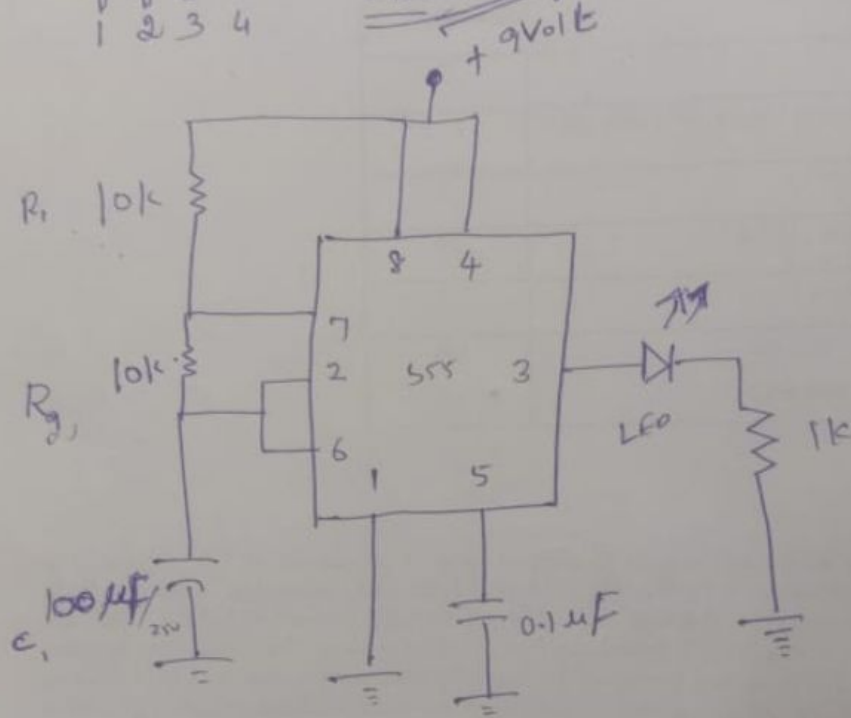
1.5 A0

Exp:2 IC 555

Pin diagram



Ckt diagram



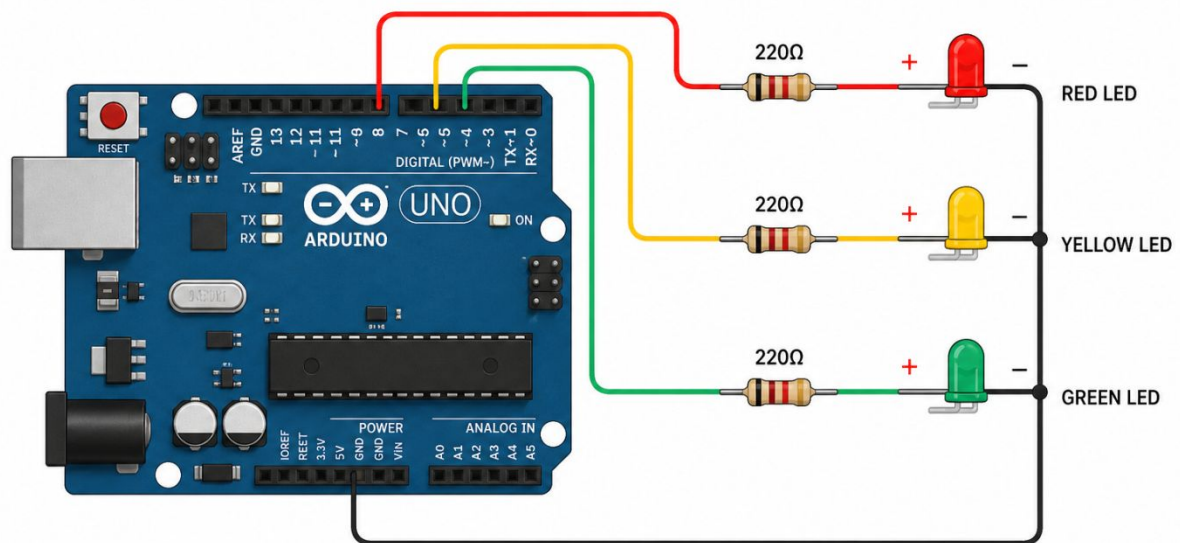
it Diagram:

no	De
	Al
	Ap
	Pr
	Re
	To

The LE

EXPERIMENT 3

TRAFFIC LIGHT USING ARDUINO UNO (WITHOUT BREADBOARD)



PROGRAM

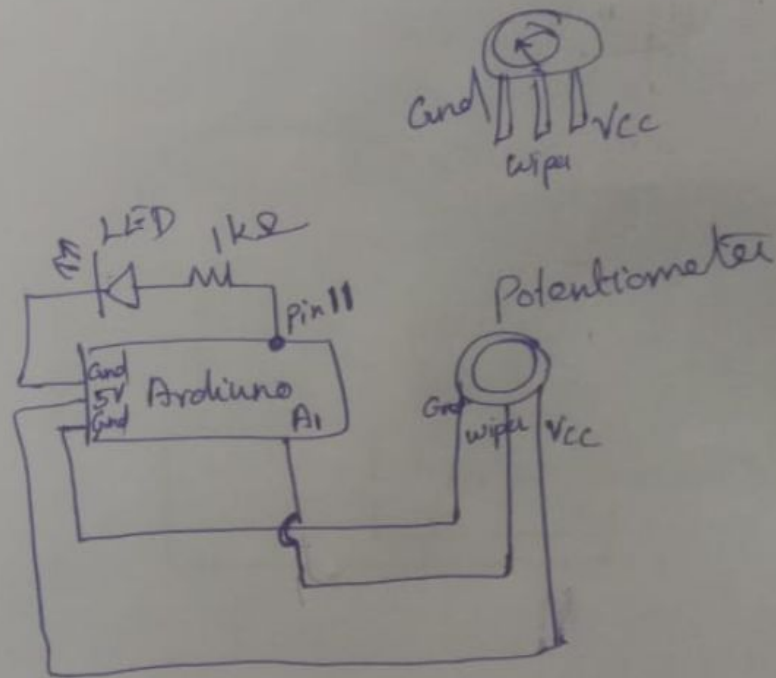
```
int red = 8;
int yellow = 9;
int green = 10;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}

void loop() {
  // RED ON
  digitalWrite(red, HIGH);
  digitalWrite(yellow, LOW);
  digitalWrite(green, LOW);
  delay(5000);
```

```
// YELLOW ON
digitalWrite(red, LOW);
digitalWrite(yellow, HIGH);
digitalWrite(green, LOW);
delay(2000);
// GREEN ON
digitalWrite(red, LOW);
digitalWrite(yellow, LOW);
digitalWrite(green, HIGH);
delay(5000);
}
```

Exp 5 Potentiometer



- Connect the extreme left (or right) leg to GND.
- Connect the other extreme leg to 5V on the Arduino.
- Add a wire between the middle pin and an analog pin.

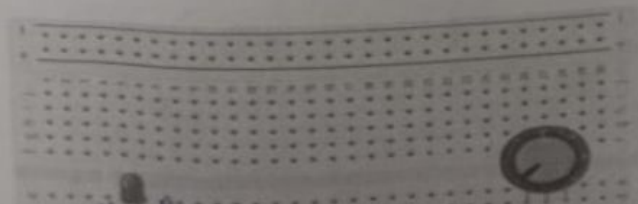
m:

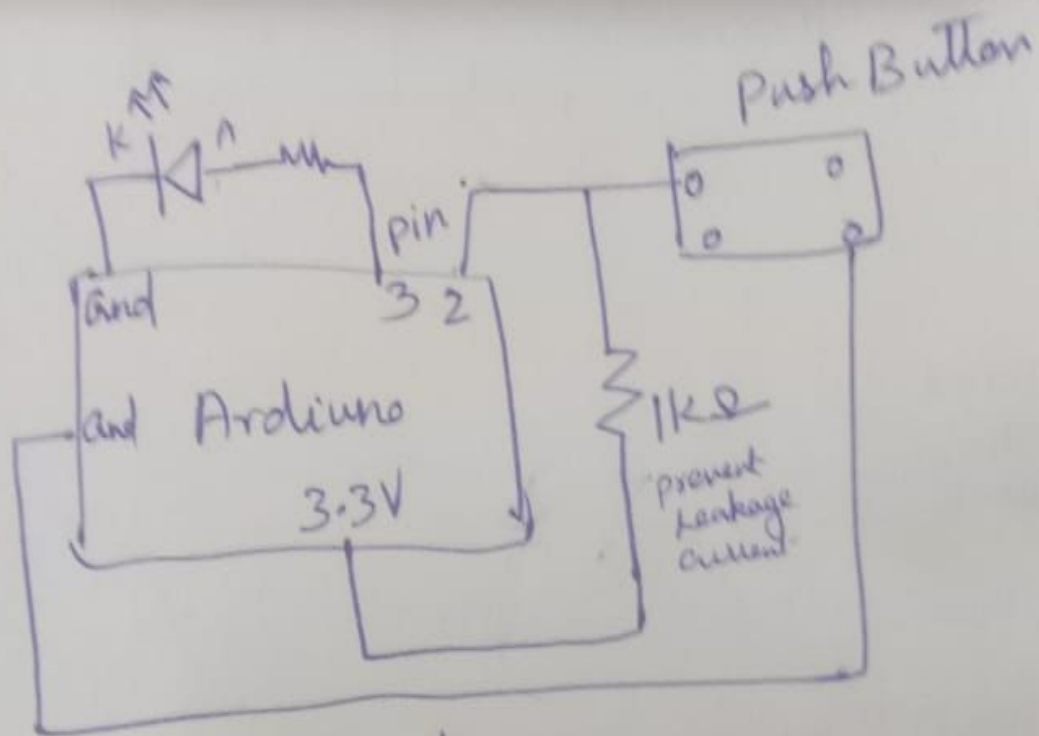
```
#define LED_PIN 11
#define POTENTIOMETER_PIN A1

void setup()
{
  pinMode(LED_PIN, OUTPUT);
}

void loop()
{
  int potentiometerValue = analogRead(POTENTIOMETER_PIN);
  int brightness = potentiometerValue / 4;
  analogWrite(LED_PIN, brightness);
}
```

OUTPUT:





Ckt diagram EX:4 LED on OFF
with push
Button

EX.NO:	4	SWITCH ON AN LED IF A BUTTON IS PRESSED
DATE:	7/8/25	

Aim: To design a simple circuit that switches on an LED when a button is pressed

Components Required:

S.no	Components	Quantity
01	Breadboard (generic)	1
02	Resistor 1k ohm	2
03	Jumper wires (generic)	few
04	Arduino UNO	1
05	LED (generic)	1
06	USB Cable	1

Platform:
Arduino IDE

Program:

```

const int BUTTON = 2;
const int LED = 3;
int BUTTONstate = 0;

void setup()
{
  pinMode(BUTTON, INPUT);
  pinMode(LED, OUTPUT);
}

void loop()
{
  BUTTONstate = digitalRead(BUTTON);
  if (BUTTONstate == HIGH)
  {
    digitalWrite(LED, HIGH);
  }
  else{
    digitalWrite(LED, LOW);
  }
}

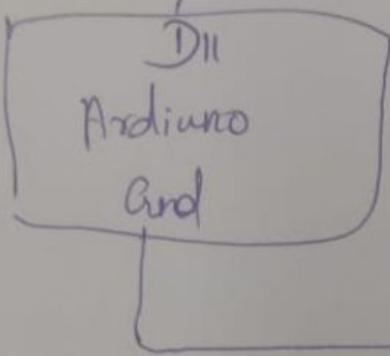
```

Ckt diagram PWM

Any pin 1k π
MANK

Aim:

To de



PWM pins: 3, 5, 6, 9, 10, 11

Platform:
Arduino

Theory:

Make the LED s
PWM (Pulse Wi
Not all pins on t



Program:

```
int LED Pin = 11; // PWM capable pin
```

```
void setup() {
```

range

```
pinMode(LED Pin, OUTPUT);  
}
```

```
void loop() {
```

```
  analogWrite(LED Pin, 128); // Run motor at 50% speed (0-255)
```

```
  delay(2000); // Run for 2 seconds
```

```
  analogWrite(LED Pin, 57);
```

```
  delay(2000); // Run for 2 seconds
```

```
  analogWrite(LED Pin, 0); // Stop motor
```

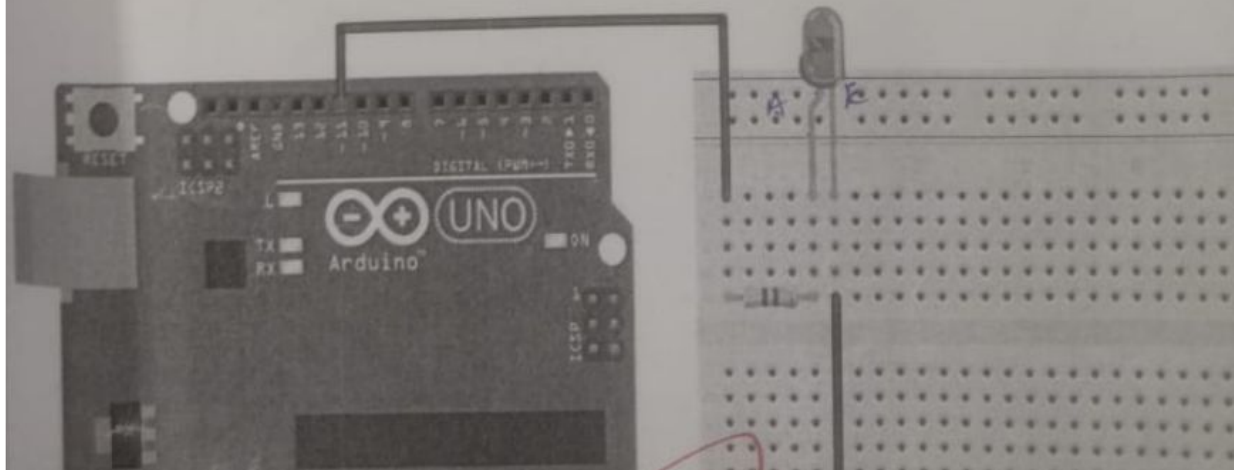
```
  delay(2000); // Wait for 2 seconds
```

```
}
```

Duty cycle 50%. (0-255)

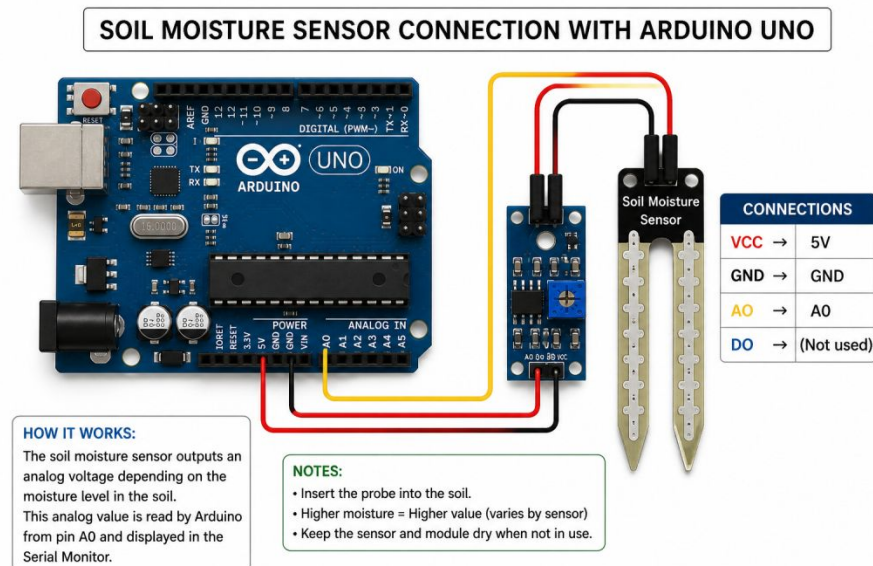
Output:

In this circuit we see that the anode of the LED is connected via a resistor to pin 11 on the cathode of the LED is connected to the ground (GND) on the Arduino



SOIL MOISTURE SENSOR

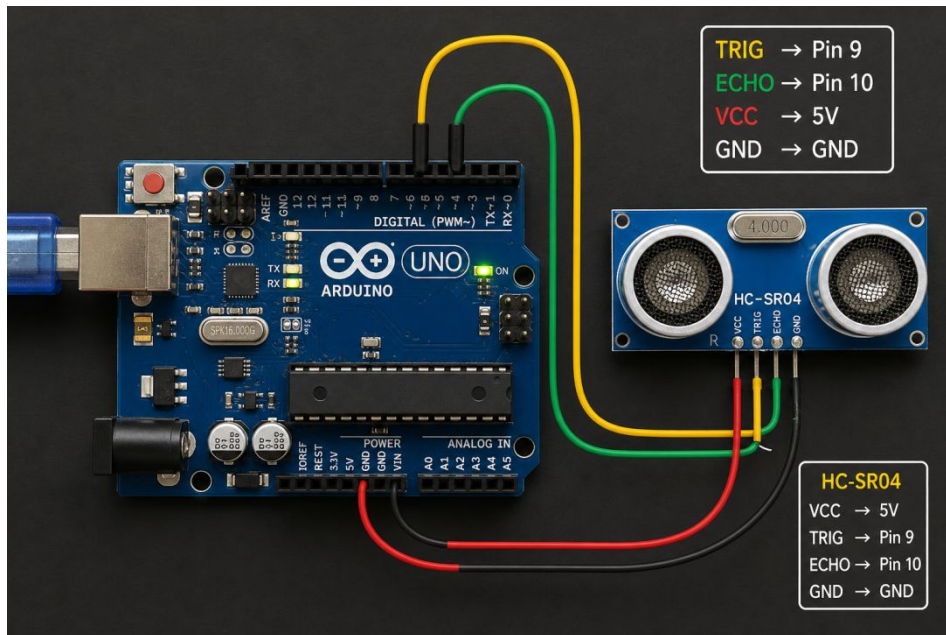
```
int sensorPin = A0; // Analog pin connected to sensor
int sensorValue = 0;
void setup() {
  Serial.begin(9600); // Start serial communication
}
void loop() {
  sensorValue = analogRead(sensorPin); // Read sensor value
  Serial.print("Soil Moisture Value: ");
  Serial.println(sensorValue);
  delay(1000); // Wait 1 second
}
```



RELAY MODULE & dc motor (LED ON OFF AND DC MOTOR SAME PROGRAM STUDY)

```
int relayPin = 7;
void setup() {
  pinMode(relayPin, OUTPUT);
}
void loop() {
  digitalWrite(relayPin, HIGH); // Relay ON
  delay(2000); // Wait 2 seconds
  digitalWrite(relayPin, LOW); // Relay OFF
  delay(2000); // Wait 2 seconds
}
```

ULTRASONIC SENSOR



```

int trigPin = 9;
int echoPin = 10;
long duration;
int distance;
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  // Clear trigger
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  // Send 10µs pulse
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  // Read echo time
  duration = pulseIn(echoPin, HIGH);

  // Calculate distance
  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);
}

```

LDR

// Define the pin where the LDR is connected

const int LDR_PIN = 34; // Use an ADC-capable pin (GPIO 34, 35, 36, 39)

void setup() {

// Initialize serial communication at 115200 baud rate

Serial.begin(115200);

pinMode(LDR_PIN, INPUT);

}

void loop() {

// Read the analog value (0 to 4095 for ESP32)

int ldrValue = analogRead(LDR_PIN);

// Print the value to the Serial Monitor

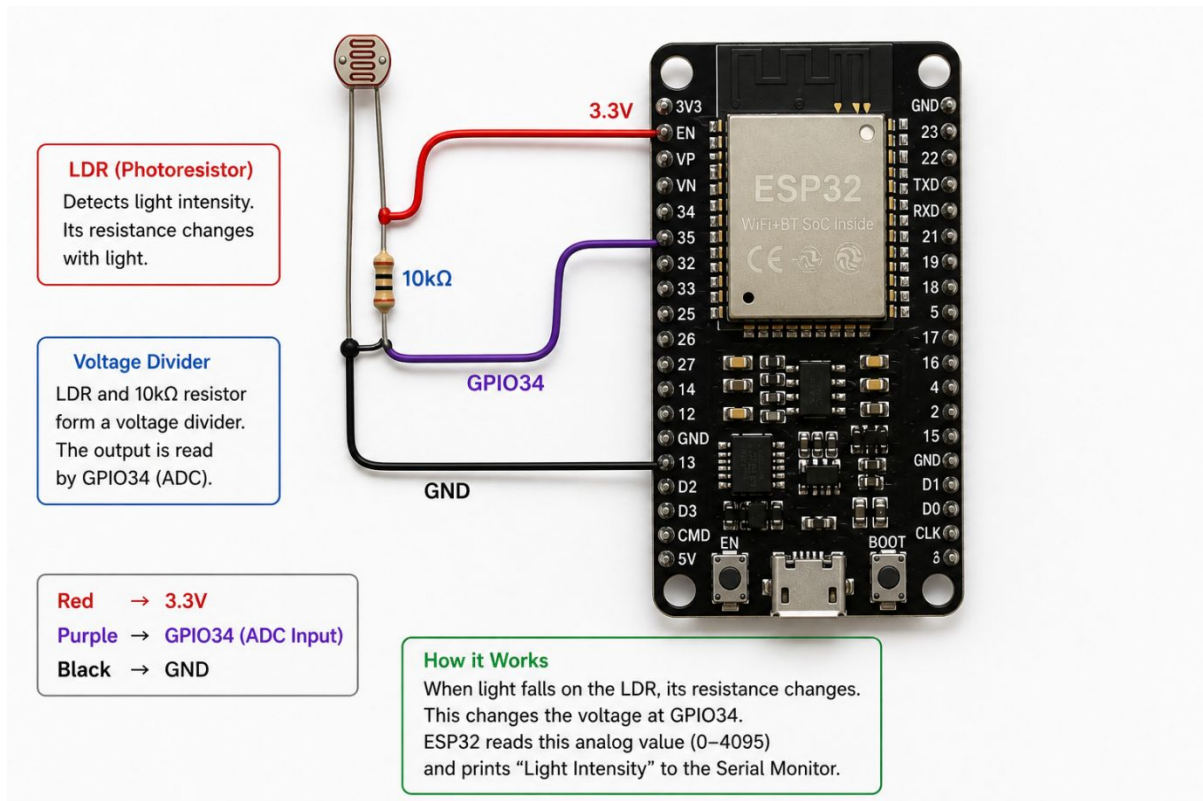
Serial.print("Light Intensity: ");

Serial.println(ldrValue);

// Small delay to make the output readable

delay(500);

}



// Water Level Sensor connected to A0

```
const int waterSensorPin = A0;  
int waterLevel = 0;
```

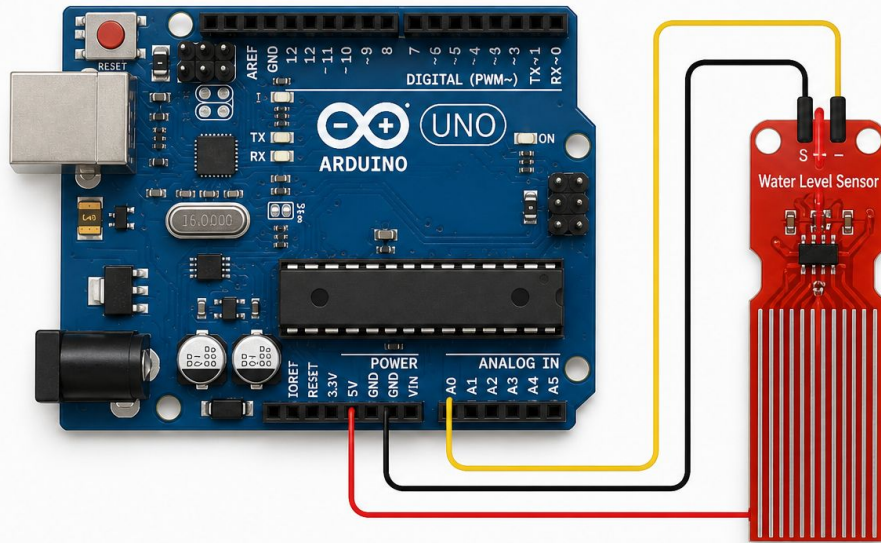
```
void setup() {  
  Serial.begin(9600);  
}
```

```
void loop() {  
  waterLevel = analogRead(waterSensorPin);
```

```
  Serial.print("Water Level Value: ");  
  Serial.println(waterLevel);
```

```
  delay(1000);  
}
```

WATER LEVEL SENSOR CONNECTION WITH ARDUINO UNO



NOTE:

Place the sensor in water. As the water level rises, the analog value (0 - 1023) will increase.

CONNECTIONS

- S → A0
- + → 5V
- - → GND

DESCRIPTION

The water level sensor outputs an analog voltage based on the water level. This voltage is read by Arduino on pin A0.